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ENGINEERING

# SIMATS Online Programming Lab

## JAVA PROGRAMMING

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### CO3-AT1-Assignment

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#### QUESTIONS

**Q1**Banking Transaction System –  
Exception Handling  
10 marks**Q2**Hospital Patient Registration –  
User-Defined Exception  
10 marks**Q3**E-Commerce Order Processing  
– Built-in Exceptions  
10 marks**Q4**Online Food Delivery – Multiple  
Threads and Execution Order  
10 marks**Q5**Manufacturing Plant –  
Producer-Consumer Problem  
10 marks

5/5 answered

**Q4**

#### Online Food Delivery – Multiple Threads and Execution Order

10

marks

An online food-delivery application uses three threads: Order Processing, Payment Processing, and Delivery Assignment.

Develop a Java program in which these activities are represented by separate threads. Analyze the problem if all three threads are started independently.

Modify the program so that:

- Order processing completes before payment processing.
- Payment processing completes before delivery assignment.
- The program uses appropriate Java thread-management techniques.
- The output clearly shows the required execution order.

Explain the difference between simply starting all threads and explicitly controlling their execution order.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus  
Order=Pizza; Payment=UPI; Delivery=Agent A Order  
→ Payment → Delivery Normal execution

Order=Burger; Payment=Card; Delivery=Agent B  
Order → Payment → Delivery Concurrency / design  
Payment=Failed Delivery assignment should not  
proceed Exception / boundary

### Your Code

```
1 class FoodDelivery{
2     public static void main(String[] args)
3         Thread order = new Thread(()->Sys
4
5         Thread payment = new Thread(()->Sy
6
7         Thread delivery = new Thread(()->{
8         order.start();
9         order.join();
10
11         payment.start();
12         payment.join();
13
14         delivery.start();
15         delivery.join();
16     }
17 }
```



### Input

Pizza UPI AgentA

### Output

Order processing completed  
Paymentprocessing completed  
Delivery assigned

✓ Submitted

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